

Version 4 Beta — Transition Architecture

Extreme Scalability, $O(xN)$ Convergence, and Hardware Convergence

VSEPR-SIM Development Day #47

From Version 3.0.X Stabilised Kernel to Version 4 Beta Formation Engine

VSEPR-SIM Development

Version 4.0.0-beta.1 — Phase 17

Contents

1 Scope: The 3.0.X $\rightarrow$ 4.0 Transition

Development Day #47 marks the boundary between the stabilised Version 3.0.X branch and the emerging Version 4 beta kernel. The 3.0.X kernel can already:

- generate repeated structure-level outputs across multiple elemental systems and lattice classes (FCC, BCC, HCP),
- track convergence, energetics, compactness, and macro-response indicators in one pipeline,
- stream live atomistic frames via dual-port NDJSON (Phase 16),
- evaluate macro-property precursors (rigidity, ductility, thermal/electrical transport, brittleness, cohesion, fracture).

What 3.0.X cannot do:

- distinguish a merely *converged* case from a *scientifically useful* case,
- quantify configurational exploration richness,
- rank output quality for downstream training or surrogate-model use,
- correlate structural compactness against macro properties,
- scale formation to $O(xN)$ with hardware-aware partitioning.

Version 4 beta addresses all five gaps by introducing:

1. the **gamma score** γ (exploration richness),
2. the **data quality score** Q_{data} (storage/publish worthiness),
3. the **compactness score** C_{compact} (structural tightness),
4. a **Pearson correlation matrix** across all final variables,
5. an $O(xN)$ **scalable formation engine** with C++26 contracts, erroneous-behaviour trapping, and hardware-convergent partitioning.

2 Gamma Score: Controlled Randomness and Richness

The gamma score quantifies how much meaningful configurational exploration occurred during generation. It does *not* reward chaos; it rewards *landscape coverage*.

$$\boxed{\gamma = w_1 S_{\text{bead}} + w_2 S_{\text{scale}} + w_3 S_{\text{var}} + w_4 S_{\text{intensity}} + w_5 S_{\text{orbital}}} \quad (1)$$

2.1 Component Definitions

S_{bead} : Bead Resolution Response

Measures sensitivity of the final state to bead-count changes. For a system run at bead count N_b and a reference run at N'_b :

$$S_{\text{bead}} = 1 - \exp\left(-\frac{|\eta(N_b) - \eta(N'_b)|}{\sigma_\eta}\right) \quad (2)$$

where σ_η is the standard deviation of η across the history. A value near 0 means the result is insensitive to resolution; near 1 means highly resolution-dependent.

S_{scale} : Multi-Scale Consistency

Compares derived properties at two scales (local bead level and macro precursor level):

$$S_{\text{scale}} = 1 - |\rho_{\text{local}} - \rho_{\text{macro}}| \quad (3)$$

clamped to $[0, 1]$. Rewards scale-consistency; penalises divergence between local packing density and macro-inferred density.

S_{var} : Configurational Spread

The coefficient of variation of η across all beads at the final step:

$$S_{\text{var}} = \min\left(1, \frac{\sigma_{\eta}}{\bar{\eta} + \epsilon}\right), \quad \epsilon = 10^{-12} \quad (4)$$

Distinguishes a homogeneous crystal (low S_{var}) from a structurally diverse morphology (high S_{var}).

$S_{\text{intensity}}$: Informational Density

Fraction of tracked state fields that are populated and finite:

$$S_{\text{intensity}} = \frac{1}{F} \sum_{f=1}^F \mathbf{1}[\text{field}_f \neq 0 \wedge \text{field}_f \in \mathbb{R}] \quad (5)$$

where F is the total number of reportable fields in the formation record. A run that populates all 22 columns of the spreadsheet scores $S_{\text{intensity}} = 1$.

S_{orbital} : Orbital Augmentation Richness

Measures the active descriptor depth:

$$S_{\text{orbital}} = \frac{\ell_{\text{max,active}}}{\ell_{\text{max,capable}}} \quad (6)$$

where $\ell_{\text{max,active}}$ is the highest ℓ with a nonzero coefficient in the unified descriptor, and $\ell_{\text{max,capable}}$ is the system’s maximum supported ℓ .

2.2 Default Weights

w_1	w_2	w_3	w_4	w_5
0.20	0.15	0.25	0.25	0.15

The weights sum to 1, keeping $\gamma \in [0, 1]$.

3 Data Quality Score

The data quality score separates “case finished” from “case is good.”

$$Q_{\text{data}} = a_1 C_{\text{conv}} + a_2 C_{\text{energy}} + a_3 C_{\text{stability}} + a_4 C_{\text{consistency}} + a_5 C_{\text{completeness}} - a_6 P_{\text{failure}} \quad (7)$$

3.1 Component Definitions

C_{conv} : Convergence Reward

Binary: 1 if the run converged within the step budget, 0 otherwise.

C_{energy} : Energy Physicality

Rewards negative (bound) final energy and penalises positive (unbound) or zero energy:

$$C_{\text{energy}} = \sigma\left(-\frac{E_{\text{final}}}{E_{\text{ref}}}\right), \quad \sigma(x) = \frac{1}{1 + e^{-5x}} \quad (8)$$

where E_{ref} is a per-element reference energy.

$C_{\text{stability}}$: Late-Step Stability

Penalises oscillation in the final 10% of the trajectory:

$$C_{\text{stability}} = 1 - \min\left(1, \frac{\sigma_{\eta}^{\text{tail}}}{\bar{\eta}^{\text{tail}} + \epsilon}\right) \quad (9)$$

 $C_{\text{consistency}}$: Cross-Field Agreement

Measures whether macro-precursor channels agree with structural metrics:

$$C_{\text{consistency}} = 1 - \frac{1}{M} \sum_{i < j} |\hat{x}_i - \hat{x}_j| \quad (10)$$

for selected normalised field pairs $\{(\rho^*, \eta^*), (\text{rigidity}, C^*), (\text{ductility}, 1 - \text{brittleness})\}$.

 $C_{\text{completeness}}$: Descriptor Population

Same as $S_{\text{intensity}}$ from the gamma score.

 P_{failure} : Failure Penalty

$$P_{\text{failure}} = p_1 \mathbf{1}[\text{step limit hit}] + p_2 \mathbf{1}[\text{NaN in output}] + p_3 \frac{n_{\text{zero}}}{F} \quad (11)$$

3.2 Default Weights

a_1	a_2	a_3	a_4	a_5	a_6
0.25	0.20	0.15	0.15	0.15	0.10

After subtraction of the failure penalty, Q_{data} is clamped to $[0, 1]$. A score above 0.7 qualifies a case for publication or surrogate-model ingestion.

4 Structure Compactness Score

$$C_{\text{compact}} = b_1 \rho^* + b_2 \eta^* + b_3 R_{\text{coord}}^* + b_4 D_{\text{macro}}^* - b_5 \Phi_{\text{void}}^* \quad (12)$$

where the starred terms are normalised to $[0, 1]$:

 ρ^* : Effective Density

$\rho^* = \min(1, \bar{\rho}/\rho_{\text{ref}})$ where ρ_{ref} is the experimental crystal density.

 η^* : Packing Efficiency

$\eta^* = \bar{\eta}$ directly from the bead environment state (already in $[0, 1]$).

 R_{coord}^* : Coordination Regularity

$R_{\text{coord}}^* = 1 - \sigma_C/(\bar{C} + \epsilon)$ Rewards uniform coordination (crystalline) and penalises irregular coordination.

 D_{macro}^* : Macro Deformation Stiffness

$D_{\text{macro}}^* = (\text{rigidity_like} + \text{cohesion_like})/2$ from the `MacroPrecursorState`.

 Φ_{void}^* : Void Fraction Penalty

$\Phi_{\text{void}}^* = 1 - \bar{\eta}$, the complement of packing efficiency.

4.1 Default Weights

b_1	b_2	b_3	b_4	b_5
0.25	0.25	0.20	0.15	0.15

5 Correlation Matrix

5.1 Variable Set

The clean variable set for the first-pass correlation:

$$\mathbf{X} = \{\bar{\eta}, \bar{\rho}, \bar{C}, \text{rms_force}, E_{\text{final}}, \text{rigidity}, \text{ductility}, \text{cohesion}, \gamma, Q_{\text{data}}, C_{\text{compact}}\} \quad (13)$$

5.2 Pearson Correlation

For variables x_i and x_j across n systems:

$$R_{ij} = \frac{\sum_{k=1}^n (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j)}{\sqrt{\sum_{k=1}^n (x_{ik} - \bar{x}_i)^2} \sqrt{\sum_{k=1}^n (x_{jk} - \bar{x}_j)^2}} \quad (14)$$

5.3 Interpretation Targets

1. Which variables most strongly predict compactness?
2. Does compactness track stiffness or ductility?
3. Does convergence quality correlate with physically meaningful outputs?
4. Do energy and compactness move together across FCC, BCC, and HCP?
5. Are γ and Q_{data} genuinely informative or decorative?

5.4 Reference Data (Day #47 Spreadsheet)

Sym	Name	Struct	Conv	Steps	$\bar{\eta}$	$\bar{\rho}$	$\bar{C}$	E_{final}	Rigid	Duct	Coh	E_{coh} (eV)
Au	Gold	FCC	T	1698	0.279	5.201	25.66	-10370	0.165	0.454	0.474	-3.81
Ag	Silver	FCC	T	6	0.001	0	0	0.26	0	0	0	-2.95
Cu	Copper	FCC	T	5	0.001	0	0	0.23	0	0	0	-3.49
Pt	Platinum	FCC	F	6000	0	0	0	263.3	0.178	0.433	0.397	-5.84
Ni	Nickel	FCC	F	5000	0	0	0	216.6	0.178	0.398	0.452	-4.44
Al	Aluminium	FCC	T	5	0.000	0.001	0	-0.001	0.21	0	0	-3.39
Fe	Iron	BCC	T	2668	0.469	11.80	45.34	-15736	0.238	0.505	0.509	-4.28
W	Tungsten	BCC	T	2954	0.494	9.394	40.09	-33207	0.246	0.504	0.616	-8.90
Mo	Molybdenum	BCC	T	2811	0.454	9.522	40.31	-25013	0.229	0.495	0.590	-6.82
Cr	Chromium	BCC	T	2684	0.467	11.67	45.00	-15065	0.237	0.504	0.607	-4.10
Ti	Titanium	HCP	T	62	0.001	0.001	0	-0.000	0	0	0	-4.85
Co	Cobalt	HCP	F	6000	0	0	0	253.5	0.091	0.379	0.361	-4.39

Observation: BCC systems (Fe, W, Mo, Cr) converge with rich structural content; FCC stalls (Pt, Ni) or produces near-zero descriptors (Ag, Cu, Al); HCP is underdeveloped. The gamma and quality scores will make these distinctions quantitative.

6 $O(xN)$ Scalable Formation Architecture

Version 4 beta transitions the formation engine from sequential per-bead evaluation to a partitioned parallel architecture.

6.1 Computational Complexity

The current kernel evaluates N_b beads over S steps. Per step, each bead queries its local neighbourhood (k neighbours on average):

$$T_{\text{step}} = O(N_b \cdot k) \quad \Rightarrow \quad T_{\text{total}} = O(S \cdot N_b \cdot k) \quad (15)$$

For fixed k (truncated neighbour list), this is $O(S \cdot N_b)$. The target is to reduce wall-clock time by a factor x via x -way partitioning:

$$T_{\text{parallel}} = O\left(\frac{S \cdot N_b}{x}\right) + O(\log x) \text{ (reduction)} = O(xN) \quad (16)$$

where x is the number of hardware work partitions (CPU threads or GPU SMs).

6.2 Hardware-Convergent Partitioning

On the target system (i9-13900K, 32 threads; RTX 4070, 46 SMs):

Partition	Hardware	Task
P-Cores (8)	Raptor Cove 5.8 GHz	Formation evolution (FIRE/env update)
E-Cores (16)	Gracemont 4.3 GHz	Neighbour rebuild, histogram, I/O
GPU SMs (46)	AD104 2.5 GHz	Batch correlation, descriptor fitting

6.3 Domain Decomposition

Beads are assigned to spatial cells; each cell is owned by one thread. Ghost-zone exchange handles cross-cell neighbours:

$$N_{\text{cell}} = \left\lceil \frac{L_{\text{box}}}{r_{\text{cut}}} \right\rceil^3, \quad \text{beads/cell} \approx \frac{N_b}{N_{\text{cell}}} \quad (17)$$

Communication volume scales as the surface-to-volume ratio:

$$V_{\text{ghost}} \propto N_{\text{cell}}^{2/3} \cdot k \quad (18)$$

which remains subdominant for $N_b > 1000$.

7 C++26 Feature Integration

Version 4 beta adopts the following C++26 / C++23 features:

7.1 Runtime Contracts

Function signatures carry executable preconditions and postconditions:

```

1  auto compute_gamma(const FormationRecord& r) -> double
2  {
3      CONTRACT_PRE(r.converged || r.steps > 0);
4      double gamma = /* ... */;
5      CONTRACT_POST(gamma >= 0.0 && gamma <= 1.0);
6      return gamma;
7  }
```

7.2 Erroneous Behaviour Trapping

With `-ftrivial-auto-var-init=pattern`, uninitialised stack variables receive the `0xFEFEFEFE` pattern, converting silent data corruption into immediate, deterministic failure.

7.3 Native Stack Traces

Using `<stacktrace>` and `-lstdc++exp`, contract violations produce full call-chain traces without external debuggers.

7.4 Flat Matrix Storage

All matrices use flat `std::vector<double>` for cache efficiency. Row-major flat indexing: `data[i*n+j]`.

8 Implementation Plan

#	Task	Header	Dependencies
1	Formation record v4	<code>v4/formation_record.hpp</code>	Day 47 spreadsheet
2	Gamma score	<code>v4/gamma_score.hpp</code>	FormationRecord
3	Data quality score	<code>v4/data_quality.hpp</code>	FormationRecord
4	Compactness score	<code>v4/compactness.hpp</code>	MacroPrecursorState
5	Correlation matrix	<code>v4/correlation_matrix.hpp</code>	all three scores
6	Heatmap export	<code>v4/corr_heatmap.hpp</code>	CorrMatrix

9 Conclusion

Development Day #47 establishes the basis for moving VSEPR-SIM from a convergent structural kernel into a continual generation platform with measurable informational value. The shift from Version 3.0.X to Version 4 beta is not merely a version increment but a methodological upgrade: each run is now judged by convergence, variation richness, data quality, and compactness relevance.

By introducing gamma-based exploration scoring, data-quality filtering, and a correlation matrix centred on compactness, the kernel becomes better suited for large-scale structured generation, ranking, and downstream thermodynamic or materials analysis. The result is a more selective, more informative, and more scientifically useful formation engine.

A prerequisite for reliable large-scale generation is that every subsystem speaks the same data language. The vector type unification completed on Development Day #56 (see the canonical bridge architecture document, Section: *Canonical Vector Type: `vsepr::Vec3`*) is part of this migration discipline. All layers — atomistic kernel, bead system, XYZ I/O, sensor, EHD, and peptide formation — now alias to a single authoritative `vsepr::Vec3`. This removes a class of silent type-mismatch errors that would otherwise surface under the increased cross-layer communication demanded by the v4 formation engine. The Eigen adapter boundary (`to_eigen/from_eigen`) ensures that future SIMD or Eigen-backed math can be introduced without rewriting every subsystem.

Humanity occasionally stumbles into competence.